



History of punch cards

Explore the various punch cards throughout history, and how they were adapted for computers. <https://digit.in/jul20-89>



Magnetic drum memory

These magnetic drums were very simple storage devices that could store about 62 kilobytes of data. <https://digit.in/jul20-90>

equipment by relaying a 'sequence of instructions' for machines such as textile looms and pianos. The holes in the paper acted as on/off switches. Basile Bouchon invented punch cards in 1725 and Herman Hollerith further developed the technology so that the information on the cards could be properly read by machines.



Fritz Pfleumer with his magnetic tape machine (1931)

One punch card could hold about 80 characters worth of data, that is fewer characters than in a Tweet today. Although punch cards are now obsolete as a data recording solution, they are still used in voting and standardised tests today. A little over a century later, Charles Babbage proposed something known as the Analytical Engine, a calculator with moving cards that used punch cards for instructions as well as response. Hollerith made Babbage's vision a reality and the holes in the punch cards didn't just hold a sequence of instructions but also stored data that the machine could read. By 1950, punch cards became a conventional technology, especially in the US, and they were regularly used for data storage and instructing machines until the mid 80s.

While punch cards were gaining momentum, another important invention was being born – magnetic tape. A long, thin strip of plastic layered with a magnetic oxide coating, the magnetic tape had the ability to store large amounts of data (for that time) on a single reel. This was the predecessor to traditional cassette tapes. The magnetic tape was first patented in 1928, by Fritz Pfleumer. The first com-

puter to use magnetic tape for storage was the UNIVAC which sold for a whopping \$750,000. Magnetic tapes had average access times but could actually stream data very quickly once the reel was wound up to the required position. One iteration of magnetic tape, the 2010 Linear Tape-Open, supported 140 MB/s transfer rates which is comparable to some hard disk drives. Magnetic tape drives could hold about 230 KB of data.

In 1932, the magnetic drum memory was invented by Gustav Tauschek. The first magnetic drum held 48 KB of memory, which is the size of about five formatted .doc files. One drum was 16 inches long and had 40 tracks that spun at 12,500 revolutions per minute. This form of storage remained common in computing until the 50s and 60s. Now, it is used as an auxiliary storage solution. However, magnetic tape drives were still much more popular since they could store a lot more data than drum memory. Magnetic tape, due to its inexpensive nature, is sometimes used to archive data even today.

In 1948, Professor Fredrick Williams, alongside his colleagues, developed the first Random Ac-



State-of-the-art HDD from the 70s

cess Memory (RAM) for storing frequently-used programs and instructions, which increased the speed of the computer overall. An array of cathode-ray tubes, in a form of a vacuum tube, were used. These acted as on/off switches and could store 1024 bits of data.

THE TURNING POINT

Magnetic tapes finally cleared the path for hard disk drives, a PC com-

ponent that is still ubiquitous today. The HDD was first introduced by IBM in 1956, however, it wasn't quite the relatively compact, streamlined component we know today. The first HDD weighed over a ton and was the size of a refrigerator! The technology comprises rapidly rotating disks that are coated with magnetic material which store data. The data stored on an HDD can be retrieved in any order rather than sequentially. The first HDD had the capacity of 4.4 MB, which was truly revolutionary at that point in time. Over the years, HDDs were constantly improved, becoming smaller, cheaper and faster over time. Today, we have HDDs that can store over 5 TB of data.

Next came the floppy disk, now immortalised as the universal Save Icon in system UIs on computers, tablets, and smartphones. We bet if we asked teens today what an image of a floppy disk meant to them, they would undoubtedly say, "the save button?". Nevertheless, this was yet another revolutionary data storage solution developed by IBM yet again, in 1967. This was a game-changing storage device since data could be transferred from one computer to another using these disks. The very first floppy disks actually were 8 inches in size, stored about 80 KB of data and had uncovered magnetic disks, but over time plastic coverings were added to keep the device protected from dirt and scratches. They also began coming out in varying sizes including a 5.25-inch version and the more popular 3.5-inch version. IBM began selling floppy disks in 1971. The storage capacity rose from 80 KB to 1.2 MB to about 250 MB later on. By the mid-70s, floppy disks became one of widely-used forms of data storage due to its portability and affordability.

Following the success of the hard disk drive, SSDs or Solid-State drives first hit the market in 1976. Dataram introduced the world's first SSD, named the Bulk Core. The device